

Nikhilesh Vijay Chirutala

Software Engineer | Python | Backend | AI

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SUMMARY

Software Engineer with 2.8+ years of experience in software development, backend engineering, API development, application design, and production support across cloud contact center, AI, CRM, trading, and business applications. Strong hands-on experience with Python, FastAPI, REST APIs, JavaScript, React.js, Next.js, MongoDB, AWS, and cloud-based application development. Experienced in translating business requirements into software solutions, developing and integrating APIs, implementing authentication and authorization, troubleshooting complex production issues, performing root-cause analysis, and delivering reliable software enhancements. Hands-on experience with RAG-based applications and LLM integration. Comfortable working independently on technical problems, evaluating implementation approaches, collaborating across teams, and supporting software delivery from development through production.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

Software Development: Software Design, Software Development, Backend Development, API Development, REST APIs, Object-Oriented Programming, Business Logic, Application Development

Backend & Frameworks: FastAPI, Node.js, Express.js, Backend Automation, Authentication, RBAC, ACL, API Integration

Frontend: React.js, Next.js, Angular, Tailwind CSS, Bootstrap, Jinja2

AI & LLM: RAG, LLM Integration, Vector Search, Prompt Engineering, AI Application Development, Conversational AI

Databases: MongoDB, Neon DB, SQL, Data Processing

Cloud & DevOps: AWS, Amazon S3, Vercel, GitHub Actions, CI/CD, Linux, Cloud Application Development, Deployment

APIs & Integrations: REST APIs, Twilio, GitHub API, Third-Party Integrations, Postman

Testing & Engineering: Unit Testing, API Testing, Postman, Jest, Debugging, Root Cause Analysis, Production Support, Git, GitHub

WORK EXPERIENCE

Software Engineer • **CA One Tech Cloud Inc.**

Apr 2026 – Present | Bengaluru, India

- Develop and maintain production software for a cloud contact center platform, contributing to backend services, application workflows, data processing, automation, and cloud operations.
- Design and develop Python and FastAPI services and REST APIs, translating business requirements and operational needs into maintainable backend solutions.
- Analyze application and production issues, perform root-cause analysis across services and data workflows, and implement reliable software fixes and enhancements.
- Develop AWS-based solutions for application data and storage workflows, applying validation, error handling, and recovery mechanisms to improve operational reliability.
- Work independently on assigned technical problems, evaluate implementation approaches, test changes, and validate software behavior before production release.

Key Projects:

CONVOSO — Cloud Contact Center Platform

Python, FastAPI, AWS, Amazon S3, REST APIs

- Developed backend services supporting lead routing, call recording management, data processing, and cloud storage workflows.
- Designed recording-recovery workflows processing 10,000+ recordings per month and restoring missing or inconsistent recording data.
- Designed and implemented S3 storage validation and metadata synchronization to maintain consistency between application records and cloud objects.
- Built backend automation to identify, recover, organize, and synchronize recordings across application and cloud storage systems, reducing manual operational effort.
- Implemented validation, error handling, and recovery logic for failed recording operations and unexpected storage states.
- Investigated failures across backend services, cloud storage, metadata, and processing workflows to identify root causes and implement corrective solutions.
- Supported production software operations by validating processing outcomes, troubleshooting failures, and delivering fixes and feature enhancements with minimal supervision.

Software Engineer • **Mantra Technologies**

Feb 2024 – Mar 2026 | Hyderabad, India

- Worked across four production applications, contributing to software design, backend development, frontend development, API integration, testing, debugging, and production support.
- Built backend modules, REST APIs, business logic, and application workflows using Python, FastAPI, and MongoDB.
- Translated application requirements into software functionality and developed maintainable solutions across backend and frontend layers.
- Developed frontend interfaces using React.js, Next.js, Angular, JavaScript, and TypeScript, integrating user workflows with backend APIs.

- Implemented authentication, RBAC, and ACL-based authorization to enforce secure access to application resources and business workflows.
- Integrated third-party services including Twilio and GitHub APIs and supported cloud deployments across AWS and Vercel.
- Maintained GitHub Actions CI/CD workflows and contributed to testing, debugging, root-cause analysis, bug fixing, deployment validation, and production issue resolution.

Key Projects:

AI Bull — AI-Powered Trading & Investment Analysis Platform

React.js, Next.js, FastAPI, Python, MongoDB, RAG, LLM

- Developed an AI-powered software platform for stock and mutual fund analysis, screening, financial data processing, and back-testing.
- Designed and implemented a Retrieval-Augmented Generation (RAG) chatbot that retrieves relevant information and generates contextual responses for application users.
- Integrated conversational AI capabilities into application workflows to provide contextual access to financial and application-specific information.
- Developed FastAPI and Python backend services, APIs, business logic, and data-processing workflows supporting the platform.
- Developed configurable screening, filtering, and back-testing capabilities for financial analysis workflows.

CRM.io — AI-Driven CRM & Communication Platform

FastAPI, Python, Jinja2, MongoDB, ACL, Twilio, REST APIs

- Developed CRM software modules for Contacts, Tasks, communication workflows, and telephony operations using Python and FastAPI.
- Designed and implemented RBAC and ACL authorization to control access to CRM resources based on user permissions.
- Integrated Twilio APIs for telephony and communication workflows and developed server-rendered interfaces using Jinja2.
- Tested APIs and end-to-end application workflows using Postman, investigated defects, and implemented software fixes.

GharPe — Real Estate Management Platform

React.js, FastAPI, Python, MongoDB, REST APIs

- Developed a React.js-based administrative application for managing real estate projects, project visibility, and platform availability.
- Designed project management workflows enabling administrators to create, update, configure, and manage projects across the platform.
- Developed lead management functionality for tracking and managing customer inquiries generated through the application.
- Built builder management functionality for maintaining builder information and associating builders with respective projects.
- Developed and integrated FastAPI/Python REST APIs supporting administrative workflows, project configuration, and data operations.

PERSONAL PROJECTS

Enterprise AI Support Assistant — AI-Powered Enterprise Knowledge & Workflow

Python, FastAPI, RAG, LLM, Vector Search, REST APIs, MongoDB

Assistant

- Designed and developed an AI-powered enterprise assistant that uses Retrieval-Augmented Generation to retrieve relevant knowledge and provide contextual responses to application users.
- Developed FastAPI REST APIs for document ingestion, knowledge retrieval, conversational workflows, and application integration.
- Implemented vector-based information retrieval and prompt-driven response generation to improve contextual accuracy of AI responses.
- Designed role-aware access controls for enterprise knowledge, allowing responses to be constrained according to user permissions.
- Structured the application as modular backend services with validation, error handling, logging, and testable API workflows.

JS Studio — Online JavaScript Playground

Next.js, GitHub API, Vercel

- Developed an online JavaScript playground for writing and executing JavaScript through a web interface.
- Integrated GitHub APIs and deployed the application with Next.js and Vercel.

AnyScrum — Project Management Platform

Next.js, Neon DB, Clerk, Vercel

- Developed a project management platform supporting projects, tasks, and team workflows.
- Implemented authentication and user management with Clerk and persistent application data with Neon DB.

Guider — Interactive Learning Portal

Angular 19, TypeScript, SSR, Vercel

- Built an interactive learning platform using Angular 19, TypeScript, and server-side rendering with reusable frontend components.
- Deployed the application on Vercel with a production-ready frontend architecture.

Contact Management App — Contacts Storage Application

React.js, JavaScript, Netlify

- Developed a React-based application for creating, viewing, updating, and managing contact information.
- Built reusable UI components and deployed the application on Netlify.

EDUCATION

B.Tech — Electrical & Electronics Engineering

2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur